Remote Code Execution: Out-of-bounds read and write in derived-item and pixel-plane handling

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Remote Code Execution: Out-of-bounds read and write in derived-item and pixel-plane handling

- Advisory: GHSA-2jg2-4ch7-h545
- Severity: critical
- Published: 2026-08-25
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Affected

- `libheif`: <=1.23.1

Description

Reaching out from Meta's Product Security team. We have received an external report about a working exploit against the latest release of libheif (v1.23.1), where parsing a malformed HEIC image file could lead to malicious code being executed.

We have seen & confirmed a functional RCE proof of concept provided by the external reporter.

Our investigation into the issue uncovered 11 separate vulnerabilities in libheif. We have developed a set of patches to mitigate most of the issues discovered.

All of the discovered vulnerabilities stem from the following incorrect assumption:

> Throughout libheif, code assumes that a `HeifPixelImage`'s logical geometry (`m_width` / `m_height`) accurately describes every channel plane it holds, and that each `heif_channel` is backed by at most one plane. Neither invariant is enforced anywhere. Derived items (`iden`, and auxiliary `auxl` chains) let an attacker construct images that violate both, and roughly a dozen consumers then index planes using geometry the planes do not have.

Attaching an agent-ingestable ZIP containing:

- Details of each of the 11 vulnerabilities discovered
- A proof of concept, where applicable, bundled with each vulnerability
- A set of patches against `v1.23.1` which mitigate a subset of the discovered issues

We are submitting this report under Meta's Vulnerability Disclosure Policy.

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